

An overview of 3D Vision-Language Models

Márcus Lobo and Vitor Matias

Afonso Paiva, Jeová Farias, Tiago Novello, Moacir Ponti

Tutorial Page



Bowdoin



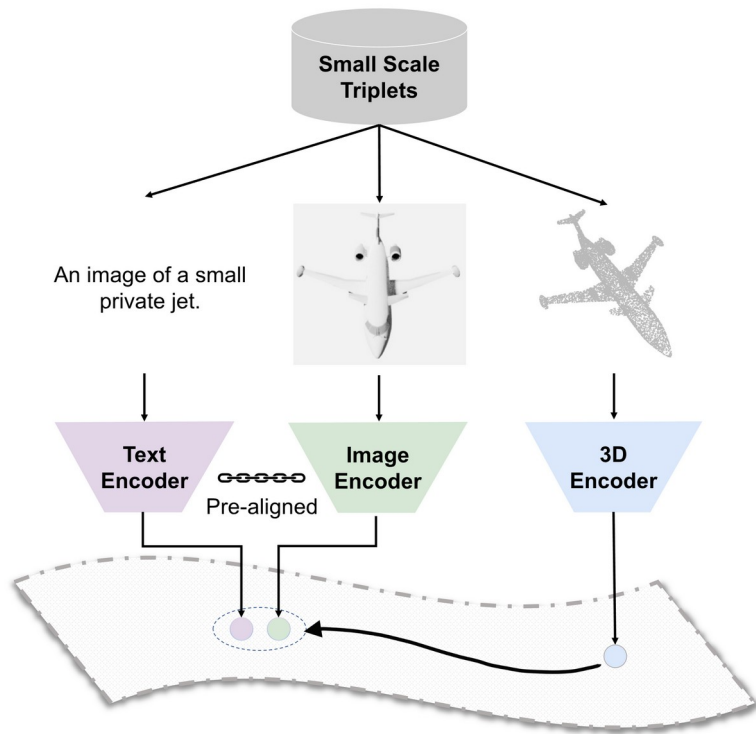
Problem Statement :: Recap

Given

- A set of triplets:
 - 3D representation, Image and Text

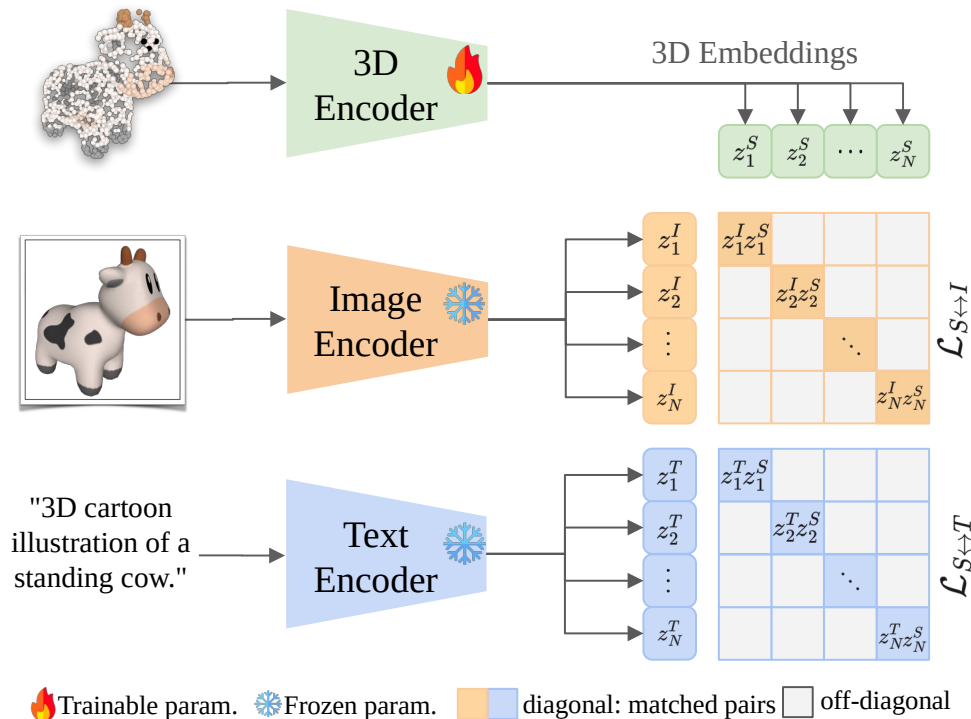
Goal

- Align the three modalities in one shared embedding space
- This is what a **3D Vision-Language Models** (3D VLMs) does.



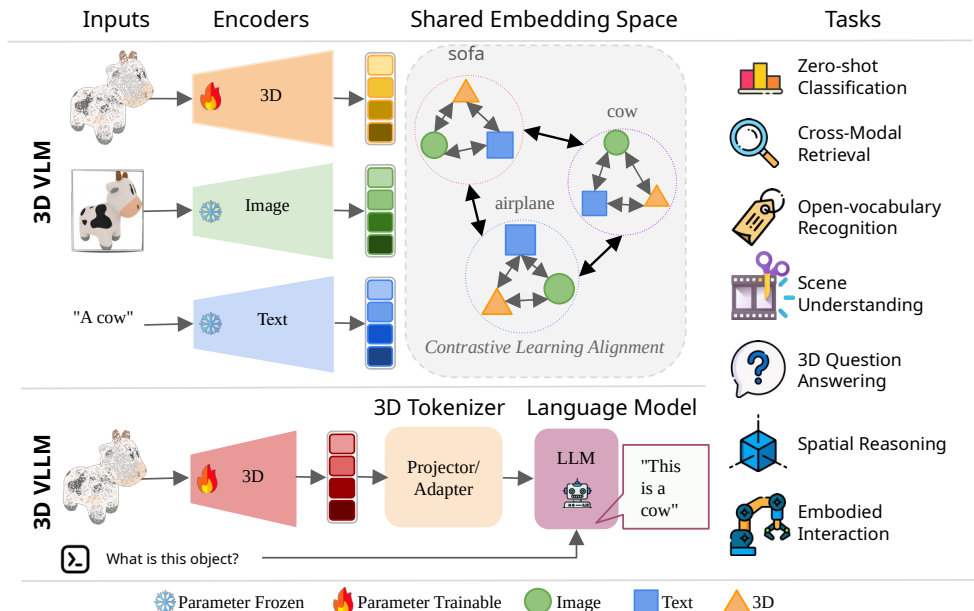
Source: ULIP framework.

3DVLMS :: Recap



- Matched triplets are the **positives** on the diagonal; off-diagonal entries are in batch **negatives**.
- Image and text encoders come from CLIP and stay **frozen**; the 3D encoder is **trained**.
- After alignment, the output **determines the downstream tasks**.

3D VLMs and 3D VLLMs :: Recap



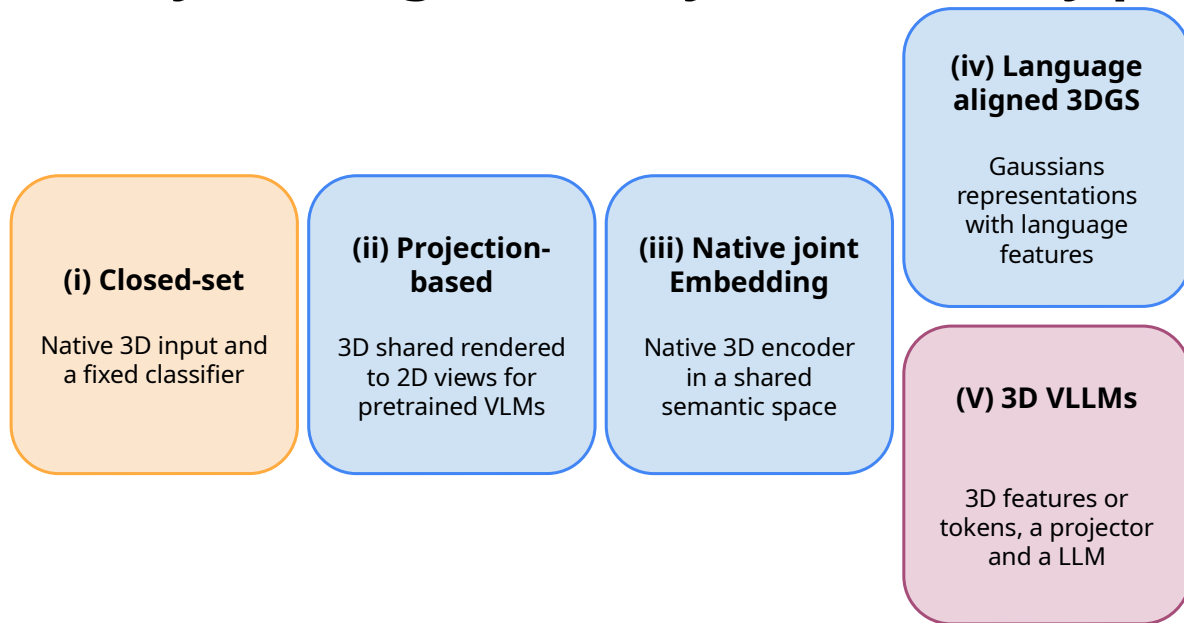
3D VLM

Compares embeddings.
Output: similarity scores.

3D VLLM

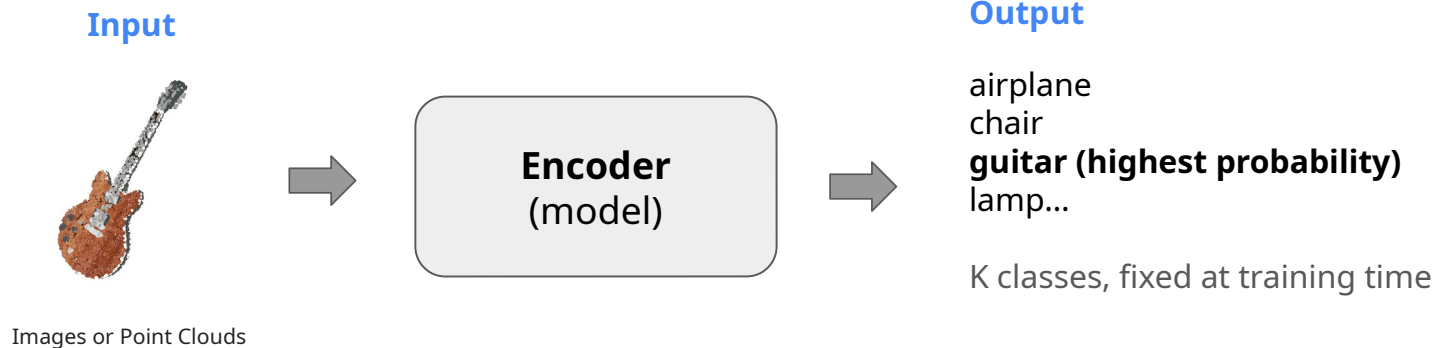
Feeds 3D tokens to an LLM.
Output: generated text.

Taxonomy, distinguished by the 3D entry point



Left to right: fixed outputs give way to open vocabularies, then to generated language.

Closed-set 3D recognition

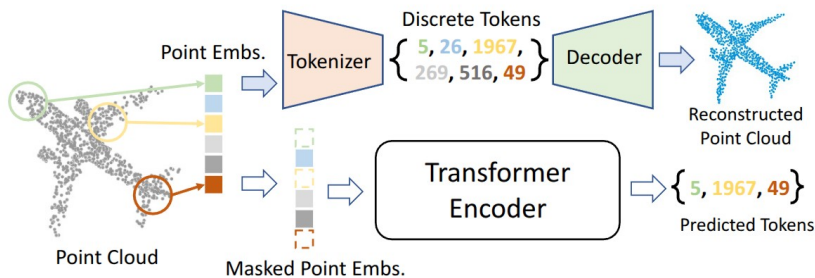


Output: $P(y|S)$ with y in the training label set.

Works: MVCNN, PointNet, DGCNN, Point-MAE, and others.

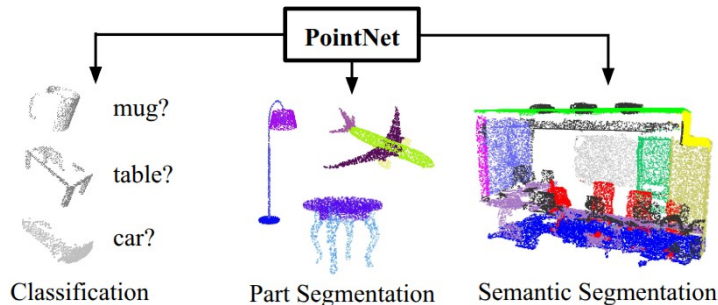
Limitation: a new phrase, e.g. “an electric guitar”, cannot be queried through the fixed classifier.

Examples from Closed-set 3D recognition



Source: Point-BERT Yu et al., CVPR 2022.

Predicts masked **geometric tokens**, not words, so it still needs a task-specific head.



Source: PointNet Qi et al., CVPR 2017.

One encoder, three tasks, and each tasks has **its own fixed label set**.

- Good encoders, **same closed space output space**. Next: connecting 3D to language.

Projection-based VLM adaptation



Output: Similarity between rendered-view of features and text.

Works: PointCLIP, PointCLIP V2, ECPL.

Trade-off: Access to pretrained 2D semantics depends on the chosen views.

PointCLIP V2: CLIP + GPT for 3D

- Combines **CLIP** and **GPT** for **open-world 3D understanding** from point clouds.
- **Visual prompting**: the point cloud is projected into more realistic depth maps.
- **Textual prompting**: GPT generates richer, 3D-specific descriptions of each.

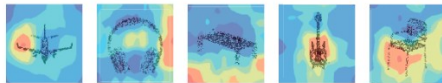
PointCLIP: Sparse Projection



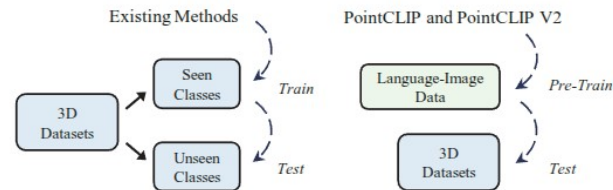
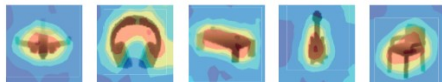
PointCLIP V2: Realistic Projection for CLIP



PointCLIP: Naive Text



PointCLIP V2: GPT Generated 3D-specific Text



Prompting GPT with 3D Command

Caption Generation Describe a depth map of a [CLASS]:	Question Answering What is a [CLASS] depth map looks like?
Paraphrase Generation Generate a synonym: A depth map of a [CLASS].	Words to Sentence Make a sentence with words: depth map, [CLASS], obscure.

3D Language Command

GPT

A grayscale map shows an airplane with wings.	A side view depth map of a cone like a pyramid.
A depth map of a chair and each pixel shows distance.	The obscure depth map shows a shallow bathtub.

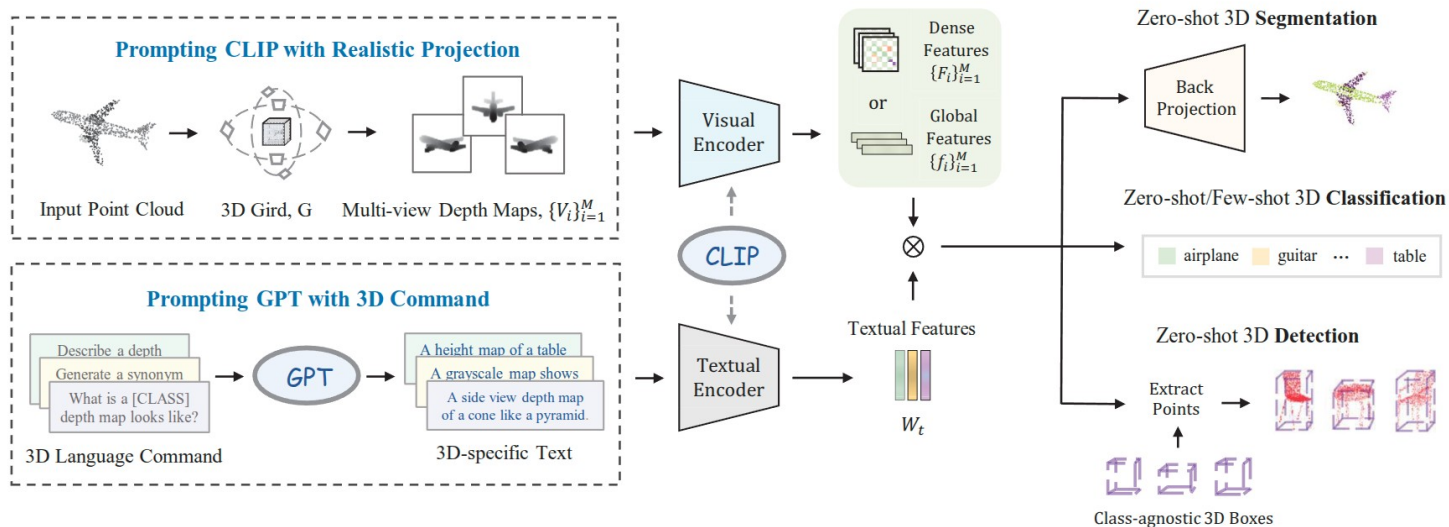
3D-specific Text

Textual Encoder

W_t

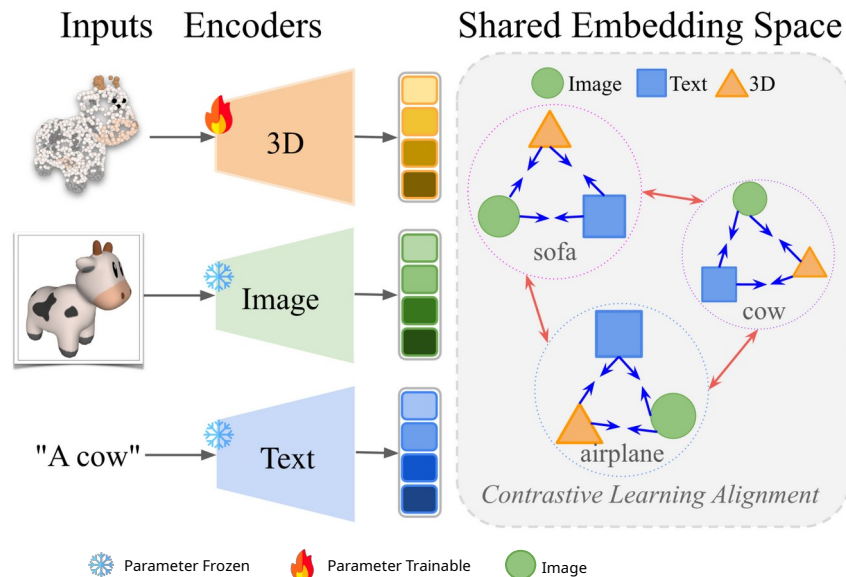
PointCLIP V2: one pipeline, several tasks

The same projection and prompts support zero-shot classification, part segmentation, and 3D detection.



Native joint embedding

- Train a dedicated 3D encoder so its features share one space with image and text.
- **Trade-off:** needs large paired datasets; supervision can be noisy and contrastive pre-training is costly.



Output: Similarity between native 3D embeddings and image/text embeddings.

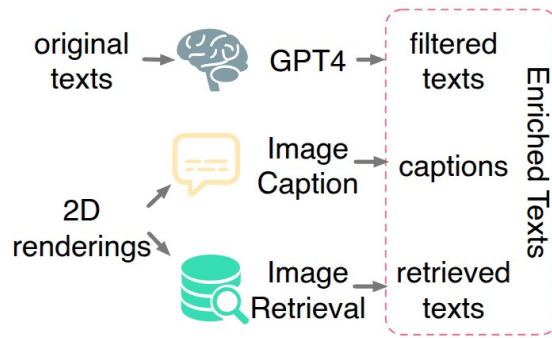
Works: CG3D, ULIP, ULIP-2, OpenShape, Uni3D, 3D-MRL, and others.

OpenShape: scaling the data

- An ensemble of 3D datasets, with GPT-4 filtering and enrich the paired texts.



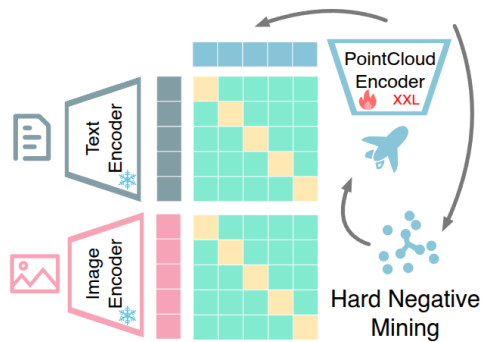
(a) Ensemble Datasets



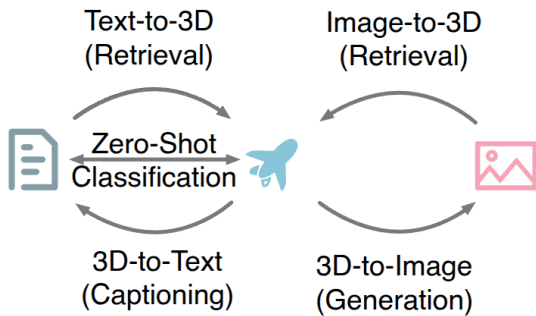
(b) Text Filtering & Enrichment

OpenShape: training and applications

- Hard negative mining sharpens the alignment.
- One embedding space then serves many applications.



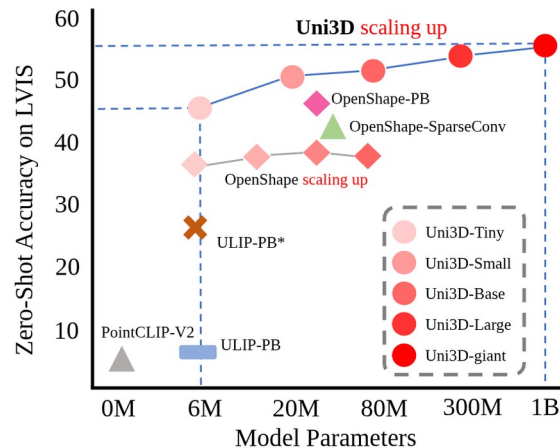
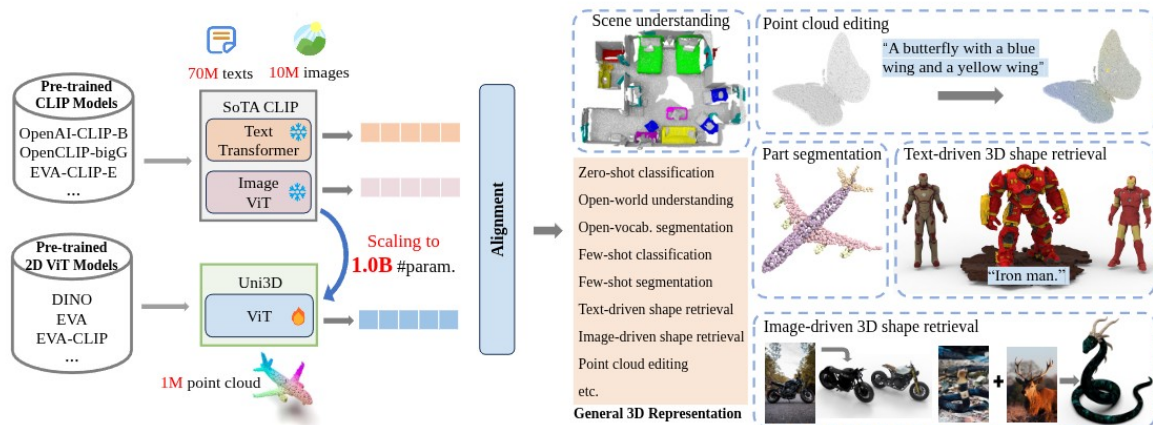
(c) Cross-Modal Alignment



(d) Cross-Modal Applications

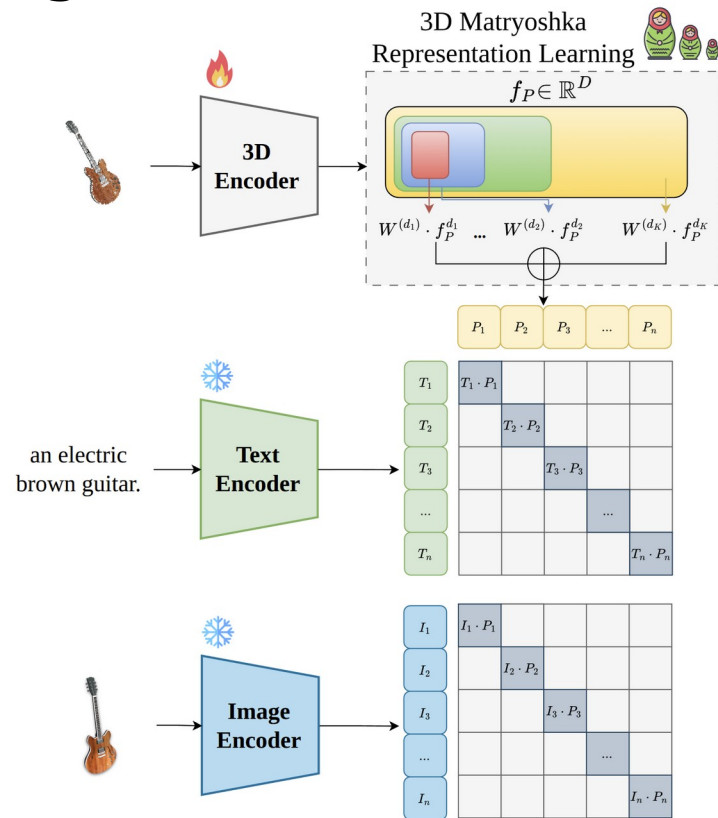
Uni3D: a 2D ViT as the 3D encoder

- A pretrained 2D ViT is adapted to point clouds and scaled up;
- Zero-shot accuracy grows with model size.

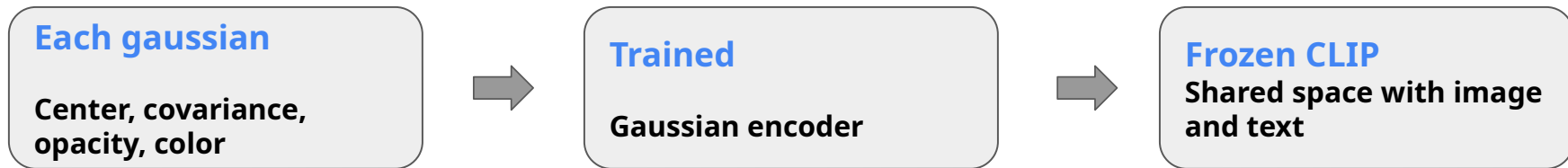


3D-MRL: nested multimodal embeddings

- A Point-BERT encoder align with frozen CLIP image and text encoders.
- The contrastive loss is applied at **several nested embeddings sizes**, Matryoshka-style.
- At inference, truncate the embeddings: smaller vectors cut storage and latency.
- One model, many embeddings sizes, no retraining.



VLMs with 3D Gaussian Splatting



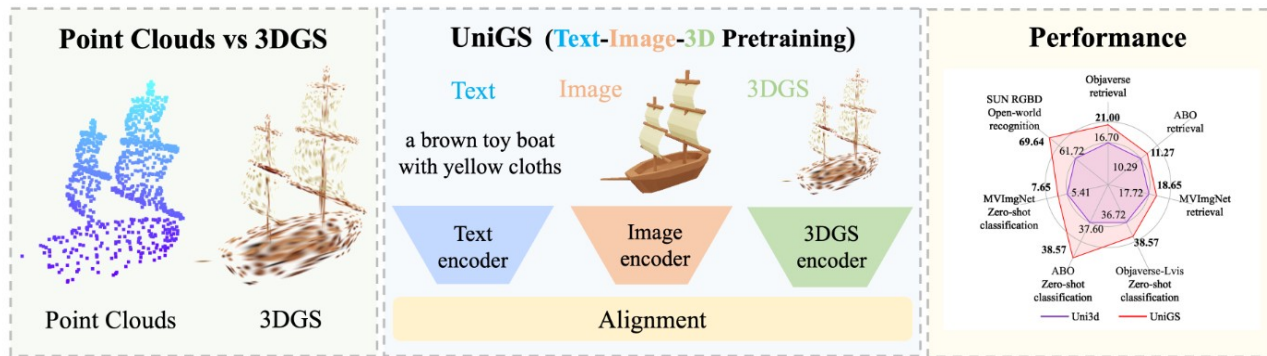
Output: Similarity between Gaussian-aware 3D embeddings and image/text embeddings.

Works: UniGS, TIGaussian, CLIP-GS, OpenGaussian, LEGaussians and others.

Trade-off: depends on reconstruction quality, can add storage and per-scene cost.

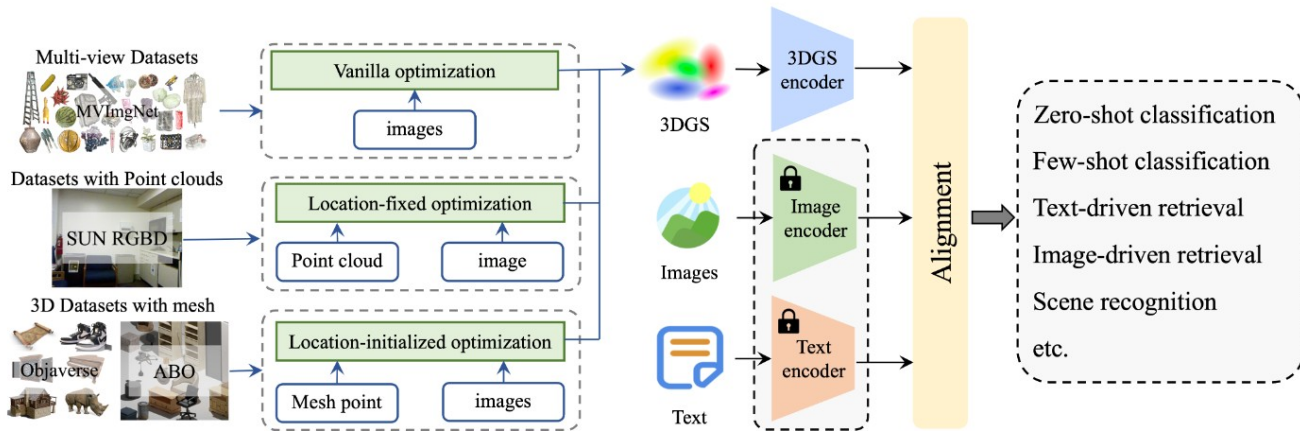
UniGS: Gaussians as the 3D modality

- Tri-modal pre-training with 3DGS in place of point clouds;
- Gaussian tokens align with CLIP image and text.

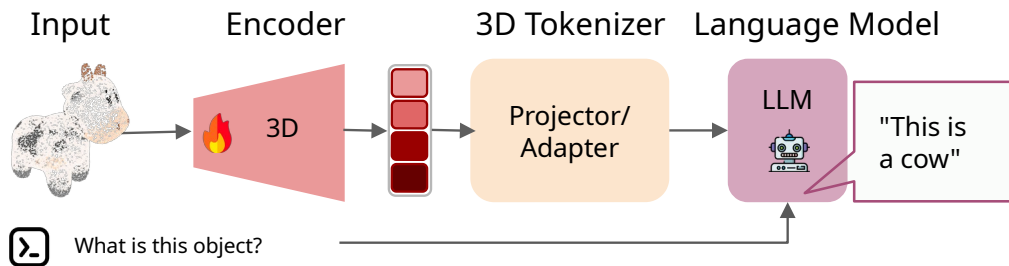


UniGS: data and downstream tasks

- Gaussian are built from multi-view;
- Point cloud and mesh datasets;
- The aligned space serves many downstream tasks.



3D Vision-Large Language Models



The objective shifts from comparing embeddings to generating language.

Recipe: 3D encoder or tokenizer, learned projector, pretrained LLM (e.g., Qwen).

Output: Text-generated from the 3D input and instructions.

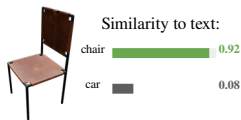
Works: PointLLM, ShapeLLM, PointAlign, Multi-3DLLM, SAGE, N3D-VLM, and others.

Trade-off: hallucination, geometric fidelity, token efficiency, limited 3D instruction data.

From model families to tasks

a) Zero-shot Classification

Prompt: "Chair" or "Table"



b) Cross-modal Retrieval

Query: "a chair"



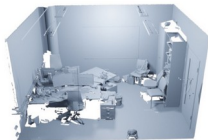
c) Visual Grounding

Prompt: "find the chair"

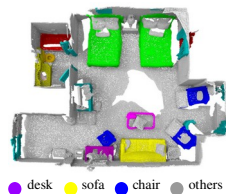


d) Questions Answering

Query: Does this scene contain a "chair"?



e) Scene Understanding



f) Spatial Reasoning

Prompt: "Where is the van in reference to the pedestrian?"



g) 3D Generation

Prompt: "draw a house"



h) Embodied AI

Prompt: "Pick up the box and put it on the table"



One aligned space (or one LLM) many tasks. They differ in what they output:

Scores: zero-shot classification

Ranked 3D assets: cross-modal retrieval

Regions: grounding, scene understanding, spatial reasoning

Text: questions answering, spatial reasoning

3D content: 3D generation

Actions: embodied AI

Task: Zero-shot classification

3D VLM

similarity scores

Name the category of a 3D shape without any labeled training example of that category.

$$\hat{y} = \operatorname{argmax}_c \cos(z^s, E_T(\text{prompt}_c))$$

HOW IT WORKS

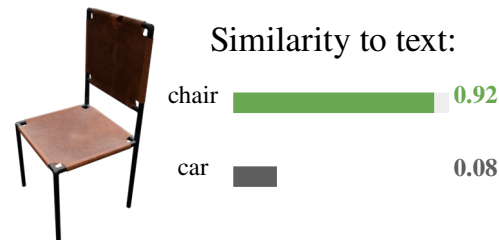
- Encode the shape once (z^s) and one text prompt per candidate class, e.g. “a point cloud of a {class}”;
- predict the class whose prompt is most similar.

EVALUATION

Top-1 accuracy on ModelNet40, ScanObjectNN and Objaverse-LVIS (the benchmark in the Uni3D scaling plot).

a) Zero-shot Classification

Prompt: “Chair” or “Table”



Task: Cross-modal retrieval

3D VLM

ranked 3D assets

Find the 3D shapes that best match a text or image query, or the text that best matches a shape.

$$\text{rank}_j \cos(z_q, z_j^s)$$

HOW IT WORKS

- Embed the whole shape gallery once and store it;
- embed the query with the frozen CLIP encoder;
- rank shapes by cosine similarity.
- It works in any direction: text $\rightarrow$ 3D, image $\rightarrow$ 3D, 3D $\rightarrow$ text.

EVALUATION

Recall@K: is the correct shape among the top-K results?

b) Cross-modal Retrieval

Query: “a chair”

Top-3 Retrieved
3D Shapes



0.94



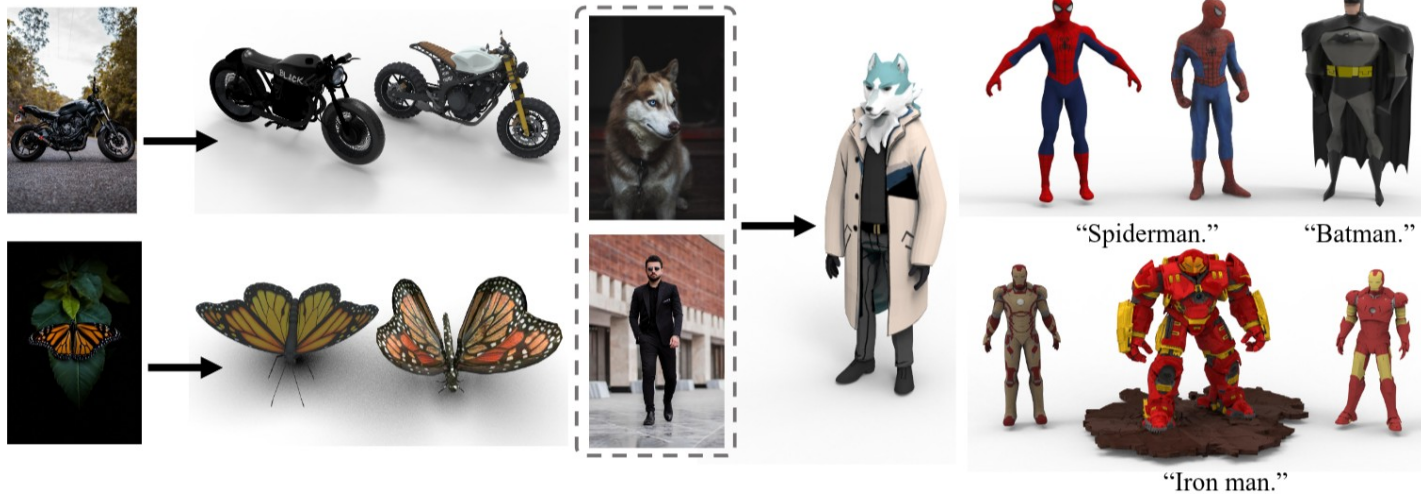
0.91



0.87

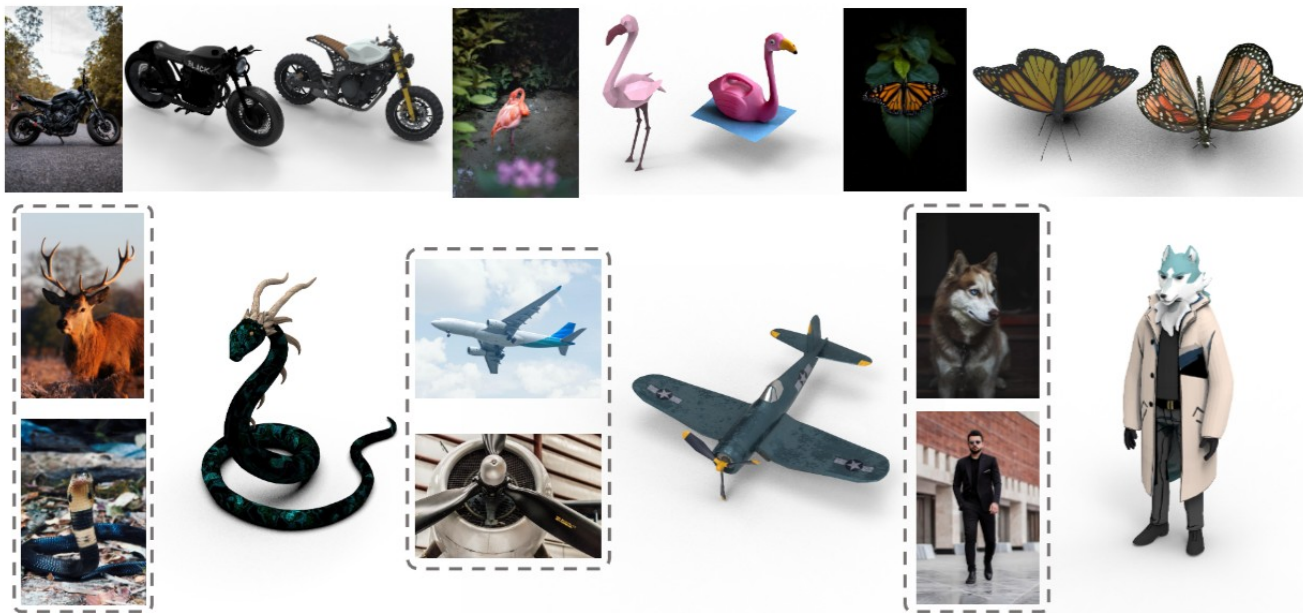
Task: Cross-modal retrieval

Image-query/text-query 3D shape retrieval results



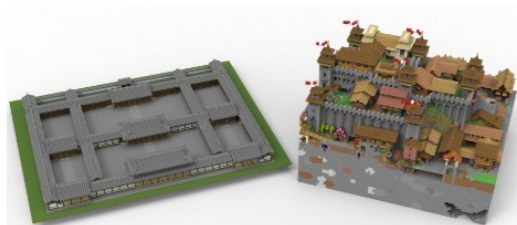
Task: Cross-modal retrieval

Image-query 3D shape retrieval



Task: Cross-modal retrieval

Text-query 3D shape retrieval



“A high quality palace.”

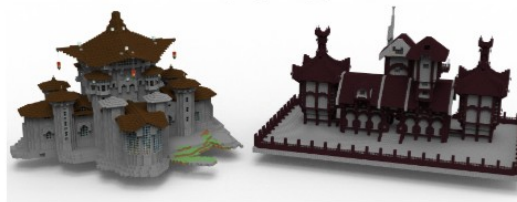


“A wooden clock.”



“Spiderman.”

“Batman.”



“A European style palace.”



“A red alarm clock.”



“Iron man.”

3D VLLMs: one recipe, many refinements

Each model keeps the encoder -> projector -> LLM recipe and fixes one weakness

Object-level foundations

PointLLM **ECCV 2024**

The reference recipe: Point-BERT
→ MLP → Vicuna, trained on 730K
Objaverse instructions.

ShapeLLM **ECCV 2024**

Stronger encoder (ReCon++),
embodied tasks and the 3D MM-
Vet benchmark.

Fixing the recipe

PointAlign **CVPR 2026**

Keeps geometry inside the LLM
with a feature-level alignment
loss.

SAGE **CVPR 2026**

No 3D encoder: point clouds
become discrete tokens, a
“foreign language”.

Beyond one object

Multi-3DLLM **CVPR 2026**

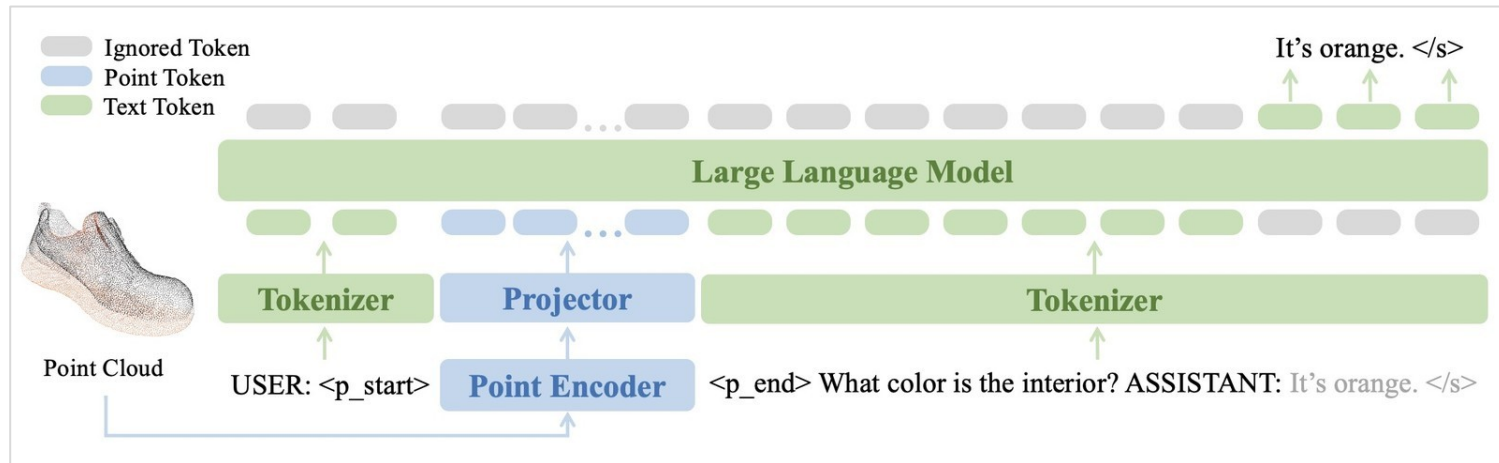
Compares 2–3 objects; new
MO3D instruction dataset.

N3D-VLM **CVPR 2026**

Grounds objects in 3D, then
reasons step by step about space.

Shared objective: next-token prediction $p(\text{answer} \mid \text{3D tokens, instruction})$

PointLLM: the reference recipe



From points to tokens

- 8,192 colored points to Point-BERT
- 513 point tokens (512 patches + [CLS])
- MLP projector.

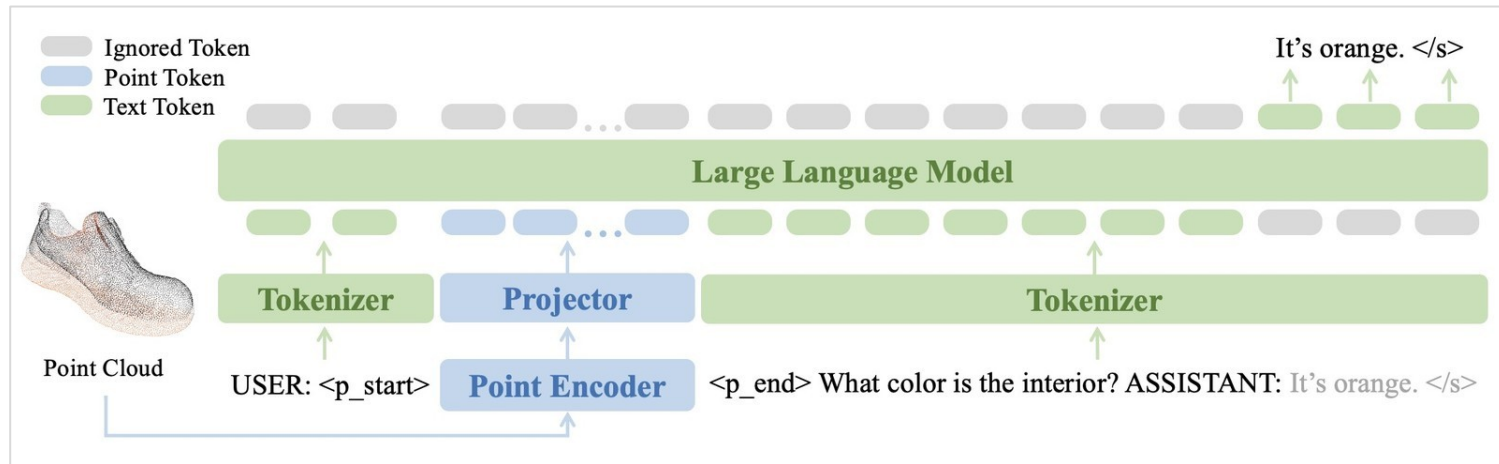
Inside the LLM

Vicuna-7B / 13B. Point tokens sit between <p_start> and <p_end> in the prompt; the loss uses only the answer tokens (gray tokens are ignored).

Data and evaluation

660K brief + 70K complex (GPT-4) instructions on Objaverse.

PointLLM: the reference recipe



{System Prompt}

USER: <p_start>{p_tokens}<p_end>{Instruction 1}

ASSISTANT: {Response 1}</s>

USER: {Instruction 2}

ASSISTANT: {Response 2}</s>

USER: {Instruction 3}

ASSISTANT: {Response 3}</s>

The **loss function is only computed on text tokens** that constitute the model's responses, including the **end-of-sentence token </s>**.

Training a 3D VLLM in two stages

The LLaVA-style recipe used by PointLLM and ShapeLLM

Stage 1: Feature alignment

Train: projector only

Frozen: point encoder + LLM

Data: 660K brief captions

Goal: point tokens that the LLM can read like words

Stage 2 : Instruction tuning

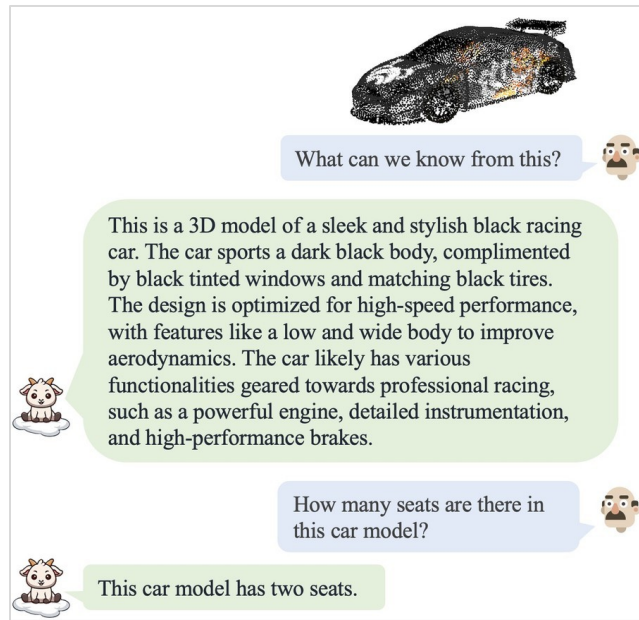
Train: projector + LLM

Frozen: point encoder

Data: 70K complex multi-turn instructions

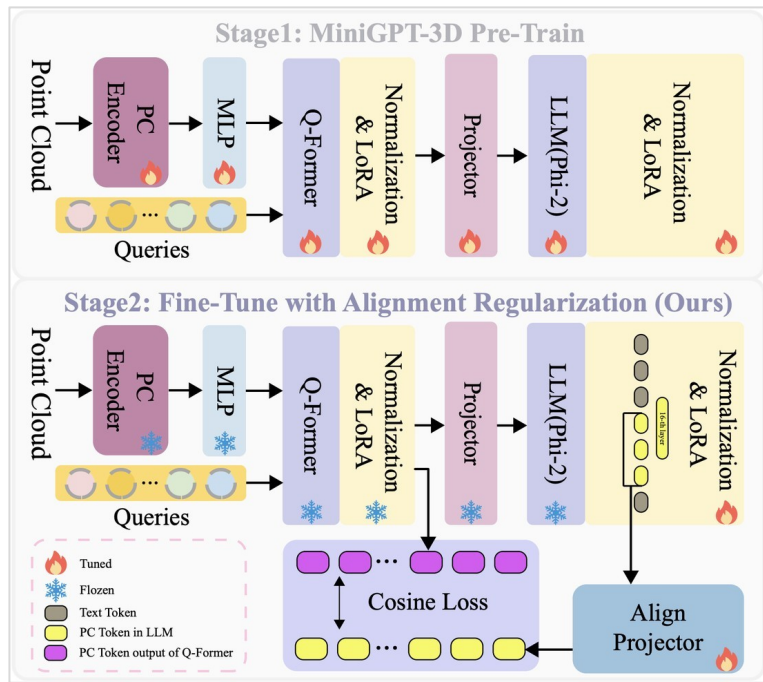
Goal: follow instructions and hold a conversation

- Next-token prediction on the answer only. The 3D encoder stays frozen in both stages.



After stage 2: a PointLLM dialogue

PointAlign: keep geometry inside the LLM

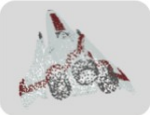


Stage 1: adopts the training recipe of MiniGPT-3D for pre-training.

Stage 2: Only trains the LoRA layers of the LLM and the alignment projector.

This alignment projector **aligns the latent representations of the point cloud tokens** in the LLM with the Q-Former output through **cosine similarity loss**.

PointAlign: some tasks

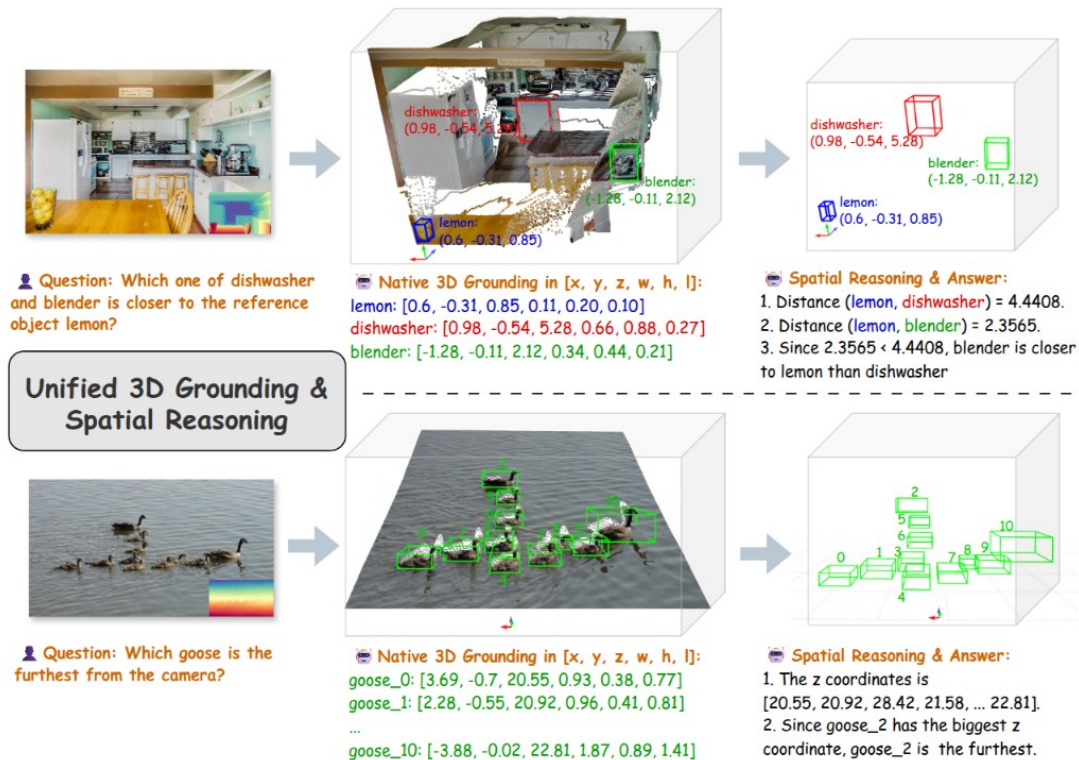
 What is this?  This is a 3D model of a cartoon-style spaceship. The spaceship is predominantly white in color, with a red stripe running along its length. It has two wings, one on each side, and a tail fin at the back.  What color is this object?  The barrel is brown in color. 3D Recognition	 What are the different components of this object?  The object is a toy car with a red body and black wheels.  Describe the object briefly.  The object is a 3D model of a dinosaur with a green body and a red mouth. It has a long neck, which adds to its unique appearance. 3D Captioning	 Provide a comprehensive description of this point cloud.  This 3D model is a simplified representation of a house, with a white exterior and brown interior. It includes a staircase leading to the upper floor, which is visible from the ground level.  How many floors does the house model have?  The house model has two floors.  What is the purpose of the staircase?  The staircase is used for moving between different levels of the house. 3D Question Answering
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PointAlign: quantitative comparison on Objaverse

Sample	
Prompt	Briefly caption this 3D model.
Ground Truth	A cartoon black monster like a dragon
InstructBLIP-13B	a black lizard with a sharp tooth in a dark room
LLaVA-13B	A 3D model of a dark, menacing dragon.
3D-LLM	A black and white tiger with long legs, standing on its hind leg.
Point-Bind LLM	The 3D model features a large, ornate gargoyle with a horned helmet, sitting on top of a building.
PointLLM-13B	The 3D model depicts a menacing black dragon, with its mouth opened wide revealing a row of sharp teeth.
MiniGPT-3D	This is a 3D model of a fearsome, black-colored dinosaur. The dinosaur possesses an aggressive stance, with sharp claws extending from its arms and legs. It has long black fangs that seem ready to attack.
PointAlign	the 3d model is a cartoon-style representation of a dragon , characterized by its large, sharp teeth and claws. the dragon is predominantly black in color, with a distinctively long tail. the dragon's mouth is open, suggesting it's in a state of attack or defense. the model is designed to be used in animations, video games, or as a decorative item.

- These improvements stem from the **partial token alignment mechanism** that enforces consistency between intermediate layer representations and Q-Former outputs during training.
- This enables the **model to retain richer point cloud semantic features** rather than discarding valuable information due to insufficient textual supervision.

N3D-VLM: ground in 3D, then reason

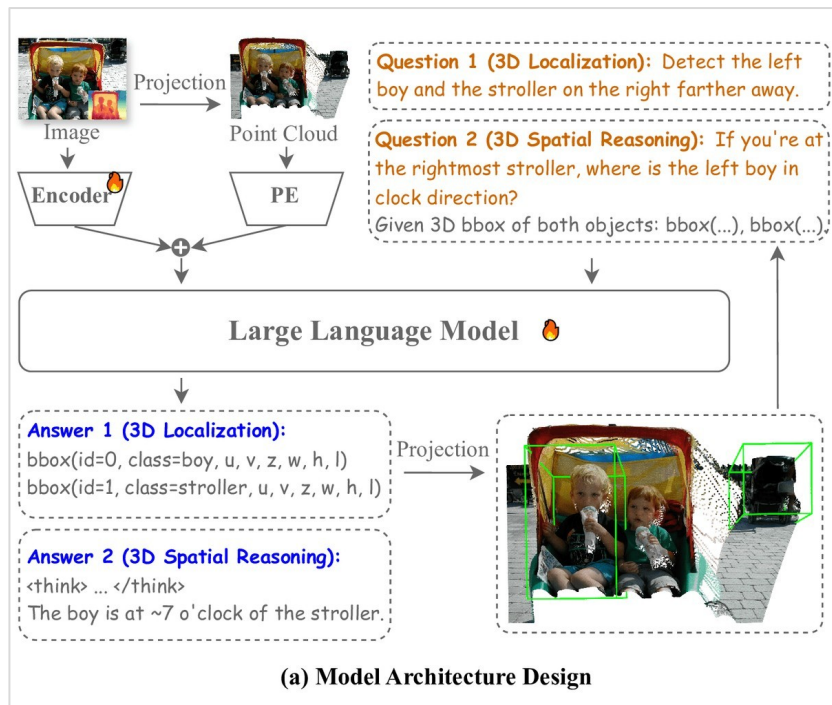


Key idea: localize objects in 3D first, then reason about their relationships.

3D grounding: predict object positions and dimensions as structured 3D bounding boxes.

Spatial reasoning: use predicted boxes to compare distances, sizes, and spatial relationships.

N3D-VLM: ground in 3D, then reason



Lift: An RGB(-D) image becomes a point cloud (monocular depth); image features + a 3D position encoding enter the LLM.

Ground: The LLM writes 3D boxes as structured text: $\text{bbox}(\text{id}, \text{class}, u, v, z, w, h, l)$.

Reason: Chain-of-thought over those boxes ($\langle \text{think} \rangle \dots \langle / \text{think} \rangle$) gives the spatial answer, e.g. “ ~ 7 o'clock of the stroller”.

Data and results

Depth estimation lifts large 2D detection datasets into 3D: $>6\times$ the largest single-image 3D detection dataset, plus spatial chain-of-thought QA.

State of the art on 3D grounding; N3D-VLM-7B beats GPT-4o, Gemini-2.5-Flash and Qwen3-VL-8B on 3D spatial reasoning benchmarks.

VLLM Task: Spatial reasoning

3D VLLM

text / regions

Answer questions about where objects are relative to each other.

HOW IT WORKS

Keep native 3D geometry (metric depth, object extent) so relations such as left/right or near/far are measured in 3D rather than guessed from pixels.

OUTPUT

A text answer, often backed by a grounded region.

f) Spatial Reasoning

Prompt: “Where is the van in reference to the pedestrian”

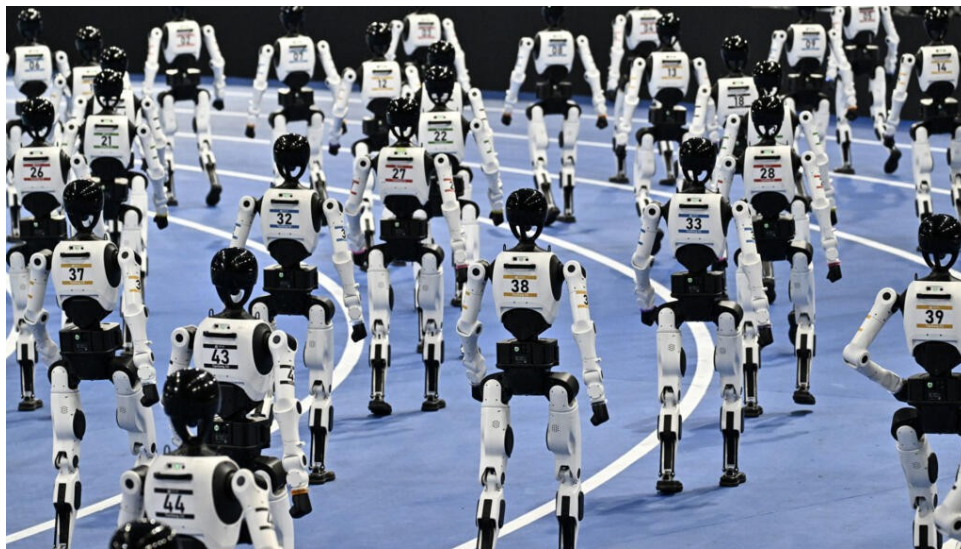


The van is on the right side of the pedestrian.

Embodied AI



Embodied AI



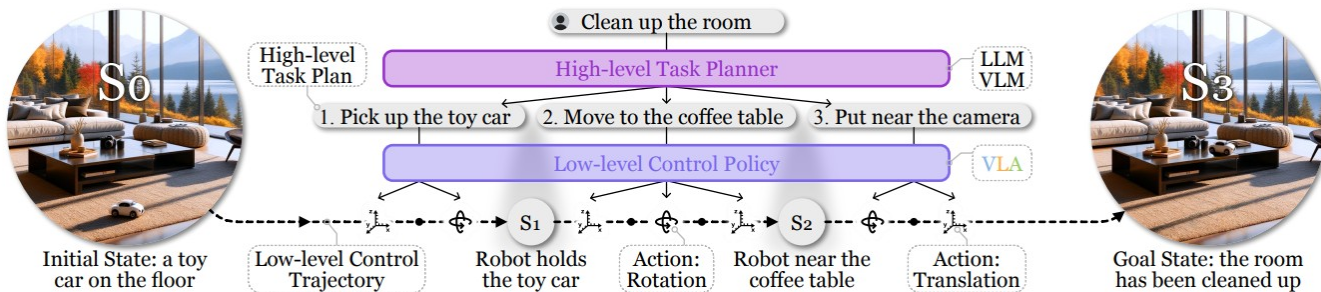
Some of these tasks were done autonomously

Embodied AI

But how?

A robot needs only

- states (s), actions (a), and rewards (r)



Embodied AI



VS



Tesla in 2018 changed from
Images + 3D to only Images

But waymo and BYD and other companies
still use

3D, Images, and even audio.

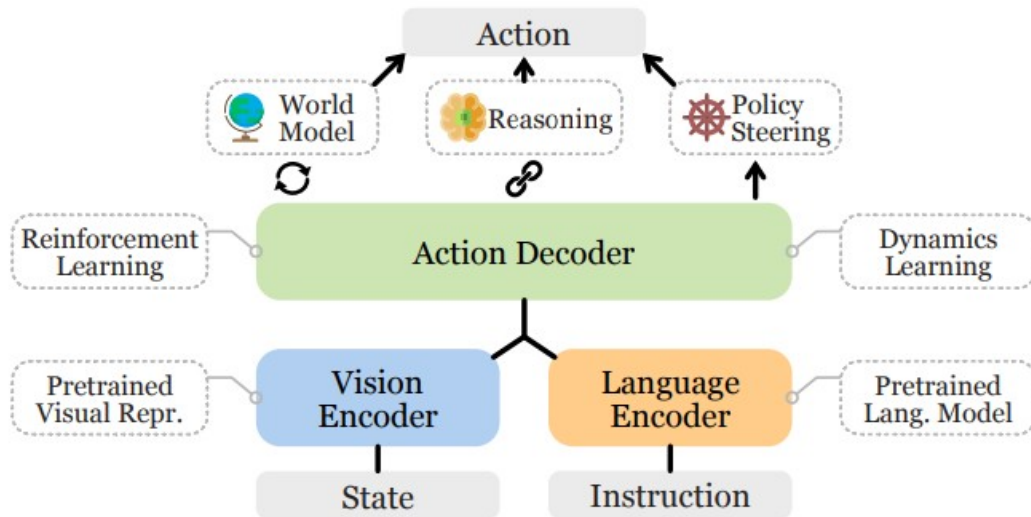
Embodied AI

But how?

A robot needs only

- states (s), actions (a), and rewards (r)

Why not 3D?



Embodied AI

A problem:

- 3D data is scarce

3D data



Images/Video/Text

Embodied AI

A problem:

- 3D data is scarce

3D data

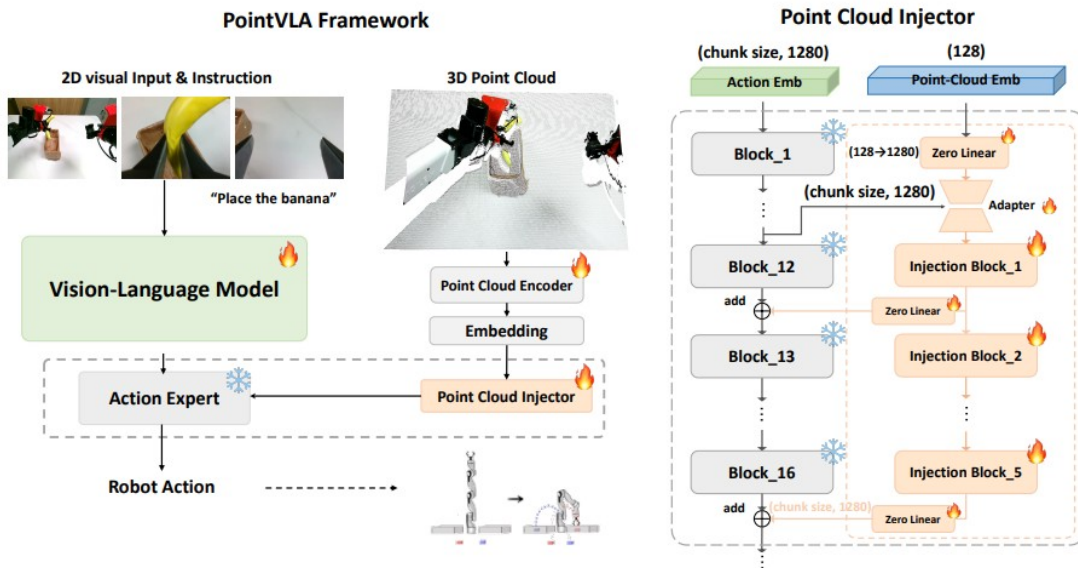


Good parts:

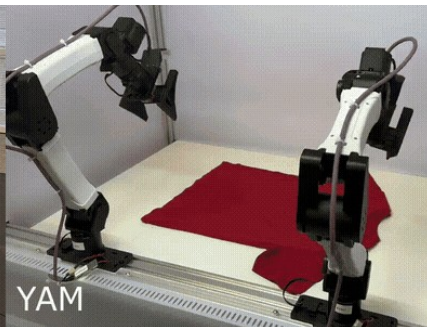
- 3D data is fast
- Accurate
- Super Cheap
 - Compute and storage

Images/Video/Text

PointVLA



Current SoTA in VLAs



Thank you :))

Márcus Lobo (<mailto:marcusvlc@usp.br>)

Vitor Matias (<mailto:vitorpmatias@usp.br>)

Afonso Paiva, Jeová Farias, Tiago Novello, Moacir Ponti

Tutorial Page



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